1 Data Generation

1.1 Pattern Matching

A *pattern* is a set.

1.1.1 Example

Let $a \circledast b = [a_1, \cdots, a_n, b_1, \cdots, b_m]$, where $a = [a_1, \cdots, a_n]$ and $b = [b_1, \cdots, b_m]$.

$$
\begin{array}{rcl}
\mathfrak{E} & := & \mathfrak{T} \cup \{\mathfrak{e}_1 \circledast \mathfrak{o} \circledast \mathfrak{e}_2 \,|\, \mathfrak{o} \in \{[\text{“}+\text{”}], [\text{“}-\text{”}]\} \wedge \mathfrak{e}_1, \mathfrak{e}_2 \in \mathfrak{E}\} \\
\mathfrak{T} & := & \mathfrak{F} \cup \{\mathfrak{t}_1 \circledast \mathfrak{o} \circledast \mathfrak{t}_2 \,|\, \mathfrak{o} \in \{[\text{“}\times\text{”}], [\text{“}\div\text{”}]\} \wedge \mathfrak{t}_1, \mathfrak{t}_2 \in \mathfrak{T}\} \\
\mathfrak{F} & := & \mathfrak{N} \cup \{[\text{“(”}] \circledast \mathfrak{e} \circledast [\text{“)”}] \,|\, \mathfrak{e} \in \mathfrak{E}\} \\
\mathfrak{N} & := & N \cup N^* \\
N & := & \{\text{“0”}, \cdots, \text{“9”}\}
\end{array}
$$

What is the difference between pattern matching and tokenwize parsing?

2 gen.jl and pattern.jl

The *map form* of a pattern $\mathfrak{P}$ is a function $\Phi' : \mathfrak{P} \to L$, where $L \subseteq \mathcal{P}(\mathfrak{P})$ is a set of *labels*[1]. The *inverse map form* of a pattern $\mathfrak{P}$ is a function $\Phi : L \to \mathcal{P}(\mathfrak{P})$, where $\Phi(l) = \{p \in \mathfrak{P} \,|\, \Phi'(p) = l\}$ for $l \in L$.

Let $G := (\langle \mathscr{G} \rangle, *)$ be a group. Let $\triangle : \Phi \times L^2 \to \Phi$!!!!!!!!!!!!WRONG, Φ should be replaced by a *set* of Φ's!!!!!!!!!!!– where Φ is the inverse map form of a pattern $\mathfrak{P} \subseteq G^{<\omega}$ and $L = \mathrm{dom}(\Phi) \subseteq G$ – be defined by $\forall l \in \mathrm{dom}(\Phi_1) \setminus \{l_1 * l_2\}$ $(\Phi_2(l) = \Phi_1(l))$ and $\Phi_2(l_1 * l_2) = \Phi_1(l_1 * l_2) \cup \{[l_1, l_2]\}$ if $l_1 * l_2 \in \mathrm{dom}(\Phi_1)$ and $\Phi_2(l_1 * l_2) = \{[l_1, l_2]\}$ if not – where $\Phi_2 = \triangle(\Phi_1, [l_1, l_2])$????????and $l_1, l_2 \in \mathrm{dom}(\Phi_1)$.

Now, consider the function $\mathscr{R} : \Phi \times \mathbb{N} \to \Phi$!!!!!!!!!defined by

$$\mathscr{R}(\Phi, n) = \begin{cases} \Phi & : \; n = 0 \\ \mathscr{R}\left(\triangle\left(\Phi, \bigcircledast_{i=1}^{2} \mathrm{Rand}\left(\mathrm{dom}\left(\Phi\right)\right)\right), n-1\right) & : \; \text{otherwise} \end{cases} .$$

Further, $\Phi_n = \mathscr{R}(\Phi, n)$, where $\Phi : \mathscr{G} \to \{\emptyset\}$ and $n \in \mathbb{N}$, is a partial or full representation of the group G, where each element in the domain is mapped thereby to the set of its neighbours.

For each $l \in \mathrm{dom}(\Phi_n)$, then, a set of vectors $\mathbf{s} \in \mathscr{G}^{<\omega}$ can be inferred with $\kappa : \Phi \times G \to \mathscr{G}^{<\omega}$!!!!!!!!! + should $\mathscr{G}$ be included in domain too? probably, yes!!!!!!!!, defined by

$$\kappa(\Phi, g) = \begin{cases} g & : \; g \in \mathscr{G} \\ \bigcup_{k \in \Phi(g)} \left(\bigcircledast_{j \in k} \kappa\left(\Phi, j\right)\right) & : \; \text{otherwise} \end{cases}$$

!!!!!!!!!!!THE DEFINITION IS WRONG, NOW THE VECTORS WILL INCLUDE SETS OF POSSIBLE VECTORS!!!!!!!!!!!

[1] Lables – or, *concepts* – are useful abstractions of patterns, which will be used later to optimize the complexity of solution-finding systems. The subsets $\mathfrak{T}, \mathfrak{F}, \mathfrak{N}$ and N in example 1.1.1 are examples of lables.